Albert Duodu

Department of Economics, Aalto University School of Business
Helsinki Graduate School of Economics (Helsinki GSE)
Ekonominaukio 1, 02150 Espoo, Finland

albert.duodu@aalto.fi
albertduodu.github.io
+358 50 401 2276

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Current Position

Postdoctoral Researcher, Aalto University and Helsinki GSE 2024–present

Education

Ph.D. in Economics, Lund University 2024
Visiting Ph.D. student: University of Warwick (host: Ludovica Gazze), Feb–Jul 2023; University of Oxford (host: Niclas Moneke), Sep 2023

M.Sc. in Economics, Lund University 2019

B.A. in Economics, University of Ghana 2014

Fields

Primary: Environmental Economics, International Trade
Secondary: Development Economics, Political Economy

References

Maria Persson
Associate Professor, Lund University
maria.persson@nek.lu.se

Joakim Gullstrand
Professor, Lund University
joakim.gullstrand@nek.lu.se

Ludovica Gazze
Associate Professor, University of Warwick
ludovica.gazze@warwick.ac.uk

Matti Liski
Professor, Aalto University and Helsinki GSE
matti.liski@aalto.fi

Zouheir El-Sahli
Economist, International Monetary Fund
zel-sahli@imf.org

Job Market Paper

Cleaner Firms, Dirtier Trade: Carbon Offshoring and the Limits of Production-Based Accounting
[PDF]

Manufacturing in rich countries is getting cleaner, at least on paper. I link plant-level fuel use for Swedish manufacturers (2006–2014) to imports weighted by their embodied carbon, and instrument firms' carbon imports with a shift-share of foreign export-supply shocks. A 10 percent rise in carbon imports lowers production-based CO_2 intensity by 6.7 percent. The decline comes from scale and product mix, not cleaner technique. The pollution-haven margin is weak on average but significant among firms with room to reallocate across products. Yet the carbon does not disappear. Domestic emissions are flat in levels, while firms' transport emissions and the international freight carrying their imports rise, and the imported goods embody 64 percent more carbon than producing them in Sweden would. Only 2 percent of imported carbon flows through foreign affiliates, and the EU's carbon border adjustment, which excludes freight, misses the transport leg. Where emissions are counted can matter as much as whether they fall.

Working Papers

Greener Under Pressure: The Local Geography of Import Competition and Emissions in Swedish Manufacturing
with Zouheir El-Sahli · *Reject and Resubmit, JAERE*
Local import competition lowers firms' CO_2 intensity through productivity gains and product-mix shifts, and the effect fades with distance between producers and importers.

The Incidence of Public Pay Reform: Earnings, Sorting, and Service Delivery under Ghana's Single Spine Policy
Revise and Resubmit, Journal of African Economies
Ghana's 2010 pay reform raised public earnings by about nine log points, almost all of it for workers with secondary education or more; health-service delivery rose less where the state employed most.

Environment and the Economy: Firm-Level Responses to Energy Price Shocks
Under review, Energy Economics
Energy tax shocks cut energy use and CO_2 but lower productivity and employment; low-productivity firms and high-skilled workers bear most of the cost.

Artisanal Mining and the Hollowing of the Formal Sector: Evidence from Ghana's Galamsey Zones
with Michael Danquah · *Under review, World Development*
Gold-suitable districts host 19 percent fewer formal firms, as the boom pulls labour into untaxed, unmeasured extraction, and annual forest loss in galamsey districts more than triples.

How Long Does a Gold Rush Last? Labour, Land and Machines after Africa's Gold Boom
Working paper
Across 33 African countries, artisanal gold work follows the world price down as well as up, but forest loss in gold-suitable places never falls back, and the 2019–24 price rise cleared more land without drawing workers back.

When the Lights Go Out: Service Failure and Political Attribution in Africa
with Maria Cedro and Beyza Gulmezoglu · *Working paper*
Across 42 African countries, outages lower presidential approval but raise protest; in South Africa, where a state-owned utility schedules load shedding, they lower satisfaction with electricity but not trust in government.

Work in Progress

- Who Bears the Carbon Border? Optimal CBAM Design with Incomplete Pass-Through (with Matti Liski)
- Lights Out, Tax Gone: The Fiscal and Institutional Consequences of Electricity Shortages (with Lennard Zyska) · *data secured*
- Electricity Rationing, Firm Dynamics and Labour Market Inequality (with Timothy Köhler and Lennard Zyska) · *data secured*
- Where Do Data Centres Go? Electricity Costs, Green Grids, and the Siting of European Computing Capacity (with Ahmed Hewidy)

Policy Papers

Aid Fragmentation and Aid Effectiveness in Sub-Saharan Africa
with B. Senadza and R. Addison · *Review of Development Finance, 2024*

Moonlighting Behavior among Migrants: Determinants and Implications for Wellbeing in South Africa
with E. Quarshie, I. Alagidede, and E.T. Tosi · *African Human Mobility Review, 2022*

Grants, Fellowships, and Awards

OP Pohjola Research Foundation, personal research grant (€16,700)	2026
Helsinki Graduate School of Economics, research grant (€6,100)	2025
Jan Wallander and Tom Hedelius Foundation, international research scholarship (SEK 250,000)	2023
Erasmus+ International Credit Mobility (€3,500)	2022
Arne Ryde Foundation scholarship (SEK 40,000)	2021
Finalist, World Econometric Game Championship, Amsterdam	2019
Swedish Institute Study Scholarship, Ministry for Foreign Affairs of Sweden	2017
Prof. Kwadwo Asenso-Okyere Prize for Best Student in Economics, University of Ghana	2015

Seminars and Conferences

2026: STEG Annual Conference; CERE Seminar (Umeå)
2025: EAERE Annual Conference (Bergen); 8th Annual LSE/Imperial Workshop on Environmental Economics; NAERE Workshop; LSE Trade Workshop; GRETA Workshop "Using Nordic Microdata to Solve Environmental Problems" (University of Helsinki); Swedish Network for European Studies in Economics and Business (SNEE); Swedish Agency for Growth Policy Analysis (Tillväxtanalys), Stockholm
2024: Yrjö Jahnsson Foundation 70th Anniversary Conference on Climate Change and Inequality (Helsinki); Majvik Ph.D. Workshop (Helsinki)
2023: Norwegian School of Economics (NHH); University of Oxford; University of Warwick Macro Workshop; Warwick–CFM–Vienna Global Macro Workshop; NOITS Annual Workshop on International Economics; SNEE; PhD Conference on Sustainable Development (Lund)
2021–22: SNEE (2021, 2022); Lund University

Teaching

Aalto University —	Lecturer, Economics of Global Challenges (B.Sc.)	2025–present
Lund University —	Lecturer, International Trade Theory (B.Sc., 3rd year)	2021–2025
	Lecturer, International Economics (B.Sc., 2nd year)	2021–2025
	Teaching Assistant, Advanced Trade Theory (M.Sc.)	2021–2025
	Supervisor, B.Sc. and M.Sc. theses	2022–2025
University of Ghana —	Teaching Assistant, Econometrics; International Trade	2015–2017

Professional Service

Referee: Environmental and Resource Economics; Empirical Economics; Journal of Productivity Analysis; African Review of Economics and Finance

Organizer: Majvik Ph.D. Workshop, Helsinki GSE, 2025–2026

Organizing committee: Swedish Network for European Studies in Economics and Business (SNEE), 2021–2023

Department: Ph.D. Admissions Committee, Department of Economics, Lund University

Data and Skills

Administrative microdata: Swedish firm, customs, energy, and employer–employee registers (Statistics Sweden, MONA); geocoded electoral and grid-cell data (PRIO-GRID, SAGE)

Software: Stata, R, Python, MATLAB, QGIS, LaTeX

Languages: English (fluent), Akan (native), Swedish (intermediate)